linearalgebra quiz1 Multi-route Grading Report

Aggregate-only supplementary dashboard; detailed thresholds and confidence intervals remain in the pairwise reports.

Scope, question, and T1 gate

Course / assessment linearalgebra / quiz1
Scope development; 6c7bf60f199a...
Population 30 submissions; 300 score rows; points
Research question On this frozen, matched split, how closely do direct multimodal and transcription-to-text grading routes agree with the same frozen reference scores?

T1 readiness passed
30 / 30 complete; 0 failed
T1 packet 3112779297d1...; run LA-v2-T1-codex-dev-30-v5_2-r2
A complete T1 route is required before either text-only G1 route can be reported.

Exact-agreement comparison

Route	Exact agreement	Value
M1		89.0%
G1-Codex		87.3%
G1-DeepSeek		84.7%

Error and risk metrics

Route	Total MAE	Within 1	Severe	Bias
M1	2.53	50.0%	46.7%	+0.16
G1-Codex	3.03	50.0%	40.0%	+0.00
G1-DeepSeek	8.63	26.7%	66.7%	−0.95

Metric definitions: exact agreement = an exact score-row match; Total MAE = mean absolute error of submission totals; Within 1 = total error at most 1 point; Severe = total error above 2 points; Bias = mean signed score-row error (route minus reference).

Reproducibility commitments

Route	Mode	Packet	Prompt	Model
M1	multimodal	d70093ab5bd4...	29123e7cd9ce...	codex_cli / gpt-5.6-sol
G1-Codex	text-only	271588f95aee...	29123e7cd9ce...	codex_cli / gpt-5.6-sol
G1-DeepSeek	text-only	271588f95aee...	29123e7cd9ce...	deepseek / deepseek-v4-pro

Canonical commitments

Snapshot SHA-256:
`6c7bf60f199a80eb27bad6cde854ce2d60eacd1fbe6995305a22226cd2797c54`
Full packet and prompt SHA-256 commitments remain in canonical public JSON at `routes[*].run.packet_hash` and `routes[*].run.prompt_hash`. The short hashes above are display locators.

Reproduce / verify

Rebuild from the seven aggregate-only inputs; the CLI flag reference is:

```
python -m benchmark.core.cli render-multi-route-report –help
typst compile report.typ report.pdf
```

Shared roster commitment: 0fe82cddead4.... Shared G1 transcript source: aec0507b6f25.... M1 and the two G1 packet hashes are intentionally retained per route rather than forced equal.

Limitations, safeguards, and operating rules

- M1 and T1 may execute in parallel because both independently consume the same frozen images; G1 begins only after T1 passes validation.
- The route contract, snapshot commitment, and full T1-to-G1 lineage binding must all match before this dashboard can be rendered.
- This report is aggregate-only and contains no student identifiers, individual scores, answers, transcripts, evidence, prompts, responses, or private paths.
- Development or calibration evidence does not by itself authorize held-out or production grading.

This dashboard is supplementary to the aggregate pairwise metric reports. Those reports retain the score unit, metric thresholds, and paired bootstrap confidence intervals for each comparison.